Read me first

FORGE ONE

What it is, the risks, installing, and getting back



FORGE ONE · Felucca 1.6.1 + FORGE 0.9 · M-VAVE FM-1
github.com/kkasemsa/FORGE-ONE (forge branch) · 22 October 2026
Felucca by Hügelton Instruments and contributors · GPL-3.0-only

FORGE ONE is custom firmware for the M-VAVE FM-1. It is Felucca 1.6.1 by Hügelton Instruments, with one addition: FORGE, a seeded generator of melodies and beats on the SEQ TOOLS F5 key, and its song starter, STARTER, on G5, and a live PERFORM mode on the FX layer. Everything else works exactly as in Felucca 1.6.1. FORGE ONE is an independent fork, not made or endorsed by Hügelton Instruments; it was called Felucca-FORGE until 0.5.1.

Based on	Felucca 1.6.1 (GPL-3.0-only) by Leo Kuroshita, Hügelton Instruments
Adds	FORGE 0.9: SEED, DENSITY, EVOLVE and 8 STYLES on SEQ TOOLS F5, melodies and beats; STARTER on G5, 45 song starters in 13 genres with the 40-sound FORGE PACK; PERFORM (FX double tap) for live mutes, solos, scenes and throws (changelog)
Branch	`forge` (`main` is an untouched copy of upstream)
Status	Builds and host tests pass in CI on every push. Installed and running on the author's FM-1 since 0.6; PERFORM (0.7) and the 45 starters (0.8) are not yet played on hardware.

Before you install

- **Installing firmware is at your own risk.** If an install is interrupted, the FM-1 can be left unable to start. You can bring it back (see If the FM-1 does not start), but that may need extra hardware.
- **Back up first.** The Felucca web editor can make a full backup of projects, presets and settings. The backup leaves out the FM6 voice bank, so keep your `.syx` file.
- **Your projects stay compatible.** FORGE only writes step data that Felucca 1.6 already stores, so projects made with FORGE open on stock Felucca 1.6 and 1.6.1.
- New to Felucca? Install stock Felucca first with the official web installer and get to know it, or try it in your browser.

Get the firmware

1. Open the Releases page and download `forge-one.fwsc` from the newest release.
2. If there is no release yet, open Actions, click the newest green **Build** run on `forge`, and download the `forge-one-…` artifact. It is a zip, and `forge-one.fwsc` is inside. You need to be signed in to GitHub to download artifacts.

Install (macOS)

The official web installer only installs stock Felucca, so install FORGE with Felucca's own command-line installer.

1. Quit Ableton Live and every other app that uses MIDI. Connect the FM-1 with a USB **data** cable and turn it on.
2. In Terminal, make a Python environment with the two MIDI libraries the installer needs. Do this once:
   - `python3 -m venv ~/fm1-env`
   - `~/fm1-env/bin/pip install mido python-rtmidi`
3. Download the installer script `tools/fm1_install.py` into the folder that holds `forge-one.fwsc`.

4. Run it from that folder: `~/fm1-env/bin/python fm1_install.py forge-one.fwsc` It asks before writing. Don't unplug the cable or let the Mac sleep until it says the install is done. The FM-1 restarts on its own.
5. Check it worked: the start-up screen says **FORGE ONE**, and when you hold **SEQ** the SEQ TOOLS map shows **FORGE** on key F5 and **STARTER** on G5.

Build it yourself (optional)

GitHub builds every push, so you only need this to build your own changes. On a Mac with Docker Desktop running:

```
git clone --recurse-submodules -b forge https://github.com/kkasemsa/FORGE-ONE.git ~/FORGE-ONE
cd ~/FORGE-ONE
tools/forge_build.sh            # set up and build build/forge-one.fwsc
tools/forge_build.sh --flash    # build and install on the FM-1
```

The first run downloads JieLi's toolchain and three AC79 SDK files and makes a Python environment in `~/fm1-env`; later runs only build. `--test` also runs the host tests (about 10 minutes). See BUILDING.md for the details.

Go back to stock

- **To stock Felucca:** use the official web installer in Chrome or Edge.
- **To the M-VAVE firmware:** use the web installer's **Return to official V15**, or M-VAVE's own updater.

If the FM-1 does not start

- **If the install stopped part-way**, run the same install command again: it finishes the write.
- **If the screen stays black**, check whether your Mac sees a USB device named **WL80UBOOT**. That is the chip's own boot mode, and the FM-1 can be recovered. The command-line installer cannot reach it:

1. Try another USB data cable, quit every MIDI app, and run the official web installer again.
2. If it stays in boot mode, FM-1 Transporter can write the flash through a Seeed XIAO RP2040 wired to the FM-1's USB lines. Back up the flash first, then write the official firmware.

The documents

Document	What it covers
Read Me First (this page)	What this is, the risks, getting it on your FM-1, getting back
FORGE User Guide (PDF)	FORGE only: controls, the 8 styles, EVOLVE, recipes
FORGE ONE User Guide (PDF)	The whole instrument: panel, pages, sequencer, engines, effects, MIDI, menu, and FORGE
Tutorial (ภาษาไทย) (PDF)	A first 4-track jam: FORGE on tracks 1–3, drums by hand on track 4

Document	What it covers
Felucca README	Upstream's own README, unchanged
BUILDING.md	Building the firmware yourself (CI does it on every push)

Credits and licence

FORGE ONE is by Khantipol (@kkasemsa): FORGE, STARTER, PERFORM and this fork's documents. It is built on Felucca:

Felucca is by Leo Kuroshita (@kurogedelic), Hügelton Instruments, and its contributors: see CREDITS.md. Its full manual is the Felucca wiki. On 16 October 2026 the maintainer stepped back from active development; Felucca stays available with occasional maintenance.

FORGE and this fork are free software under GPL-3.0-only, like Felucca. The bundled fonts, ported DSP and SDK files keep their own licences (LICENSES/, LICENSING.md). M-VAVE and FM-1 are trademarks of their respective owners. Neither Felucca nor this fork is affiliated with or endorsed by them.